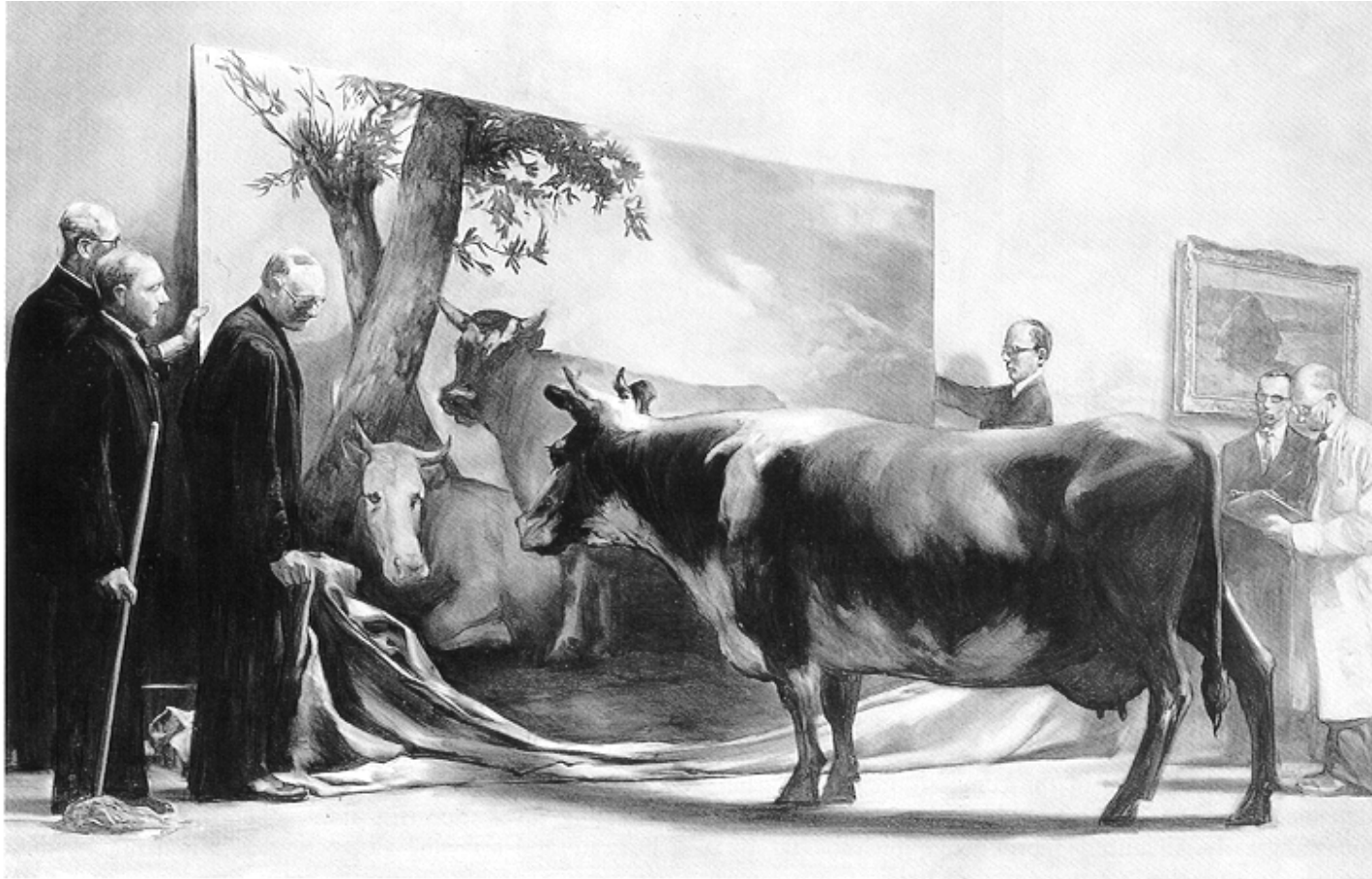


ENVX2001: Applied Statistical Methods



Lecture 3 – Distance & (Dis)similarity Measures

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Outline

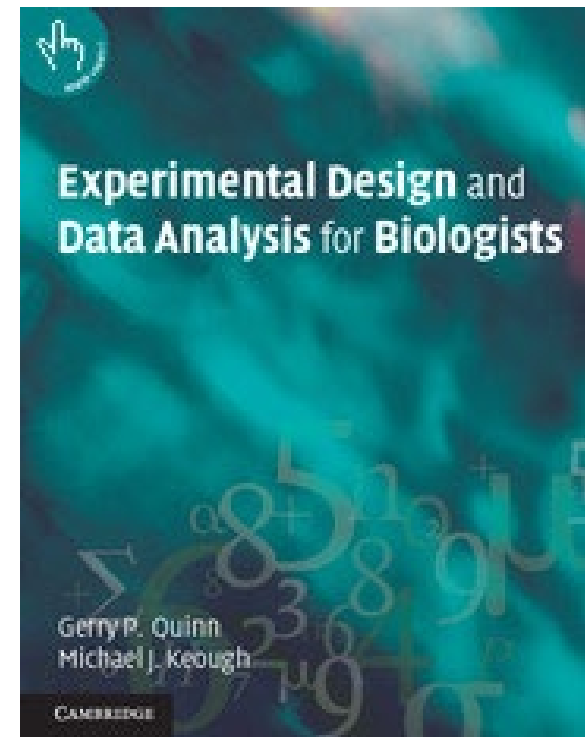
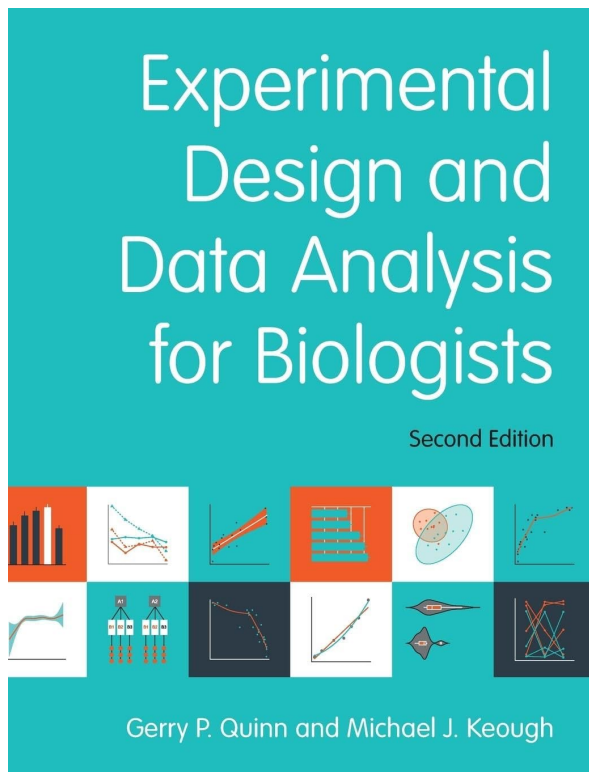
- Multivariate data and
- Hypothesis testing and data exploration*
- Important techniques including
 - PCA
 - Cluster analysis*
 - nMDS
 - Monte Carlo approaches
 - MANOVA

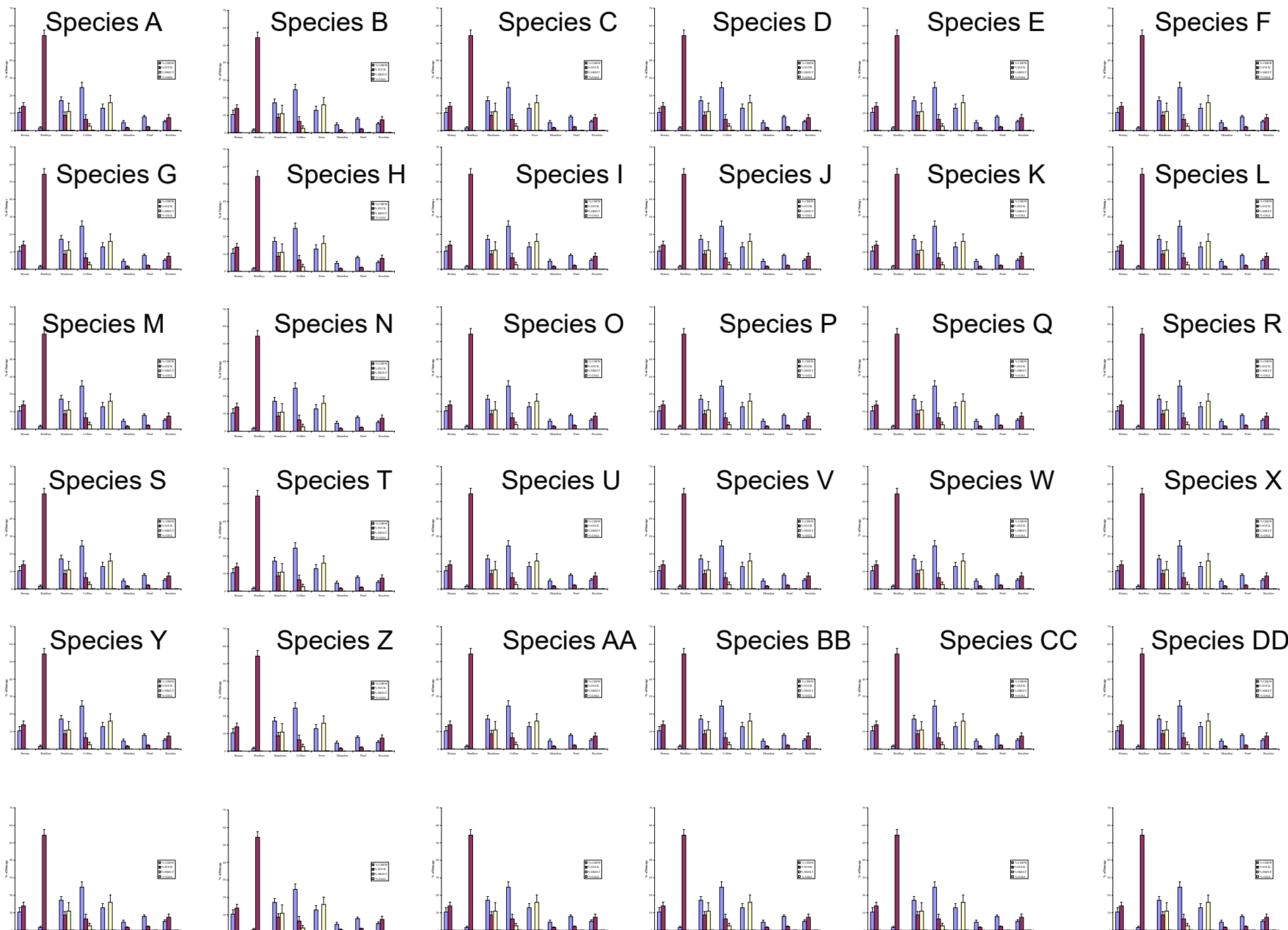
Lectures 3, 4 & 5

- Distance matrices
- Transformations and standardisations
- Cluster analysis
 - Types of questions
 - Carrying it out
 - Interpreting it
- Hypothesis testing (and best practice)
- Steps for testing
 - Visualising (nMDS)
 - Randomization (generating p using ANOSIM and PERMANOVA)
 - Post-hoc tests
 - Identifying contributors to the differences (SIMPER)

References

- Quinn & Keough (2002) – Chapter 15
- Quinn & Keough (2023) – Chapter 14,16





General principles

Distance measures

Transformation

Standardization

What we do – community structure

Three main categories of multivariate analysis that are common in ecology:

- (i) **clustering** - generally concerned with the identification of clusters or groups of observations (or species). Most often used for the generation of hypotheses in the beginning of a study or for exploratory purposes
- (ii) **ordination** – comprehensible rearrangement of elements. We want to produce an ordination in order to visualize patterns in the multivariate cloud of data (by reducing dimensions in some way)
- (iii) **statistical tests of hypotheses** - rigorously test explicit hypotheses concerning those patterns by reference to *a priori* groups or relationships with predictors, such as environmental variables

Our focus on where we have a specific hypothesis in mind and have either

- sampled natural systems in a structured way or
- done a manipulative experiment

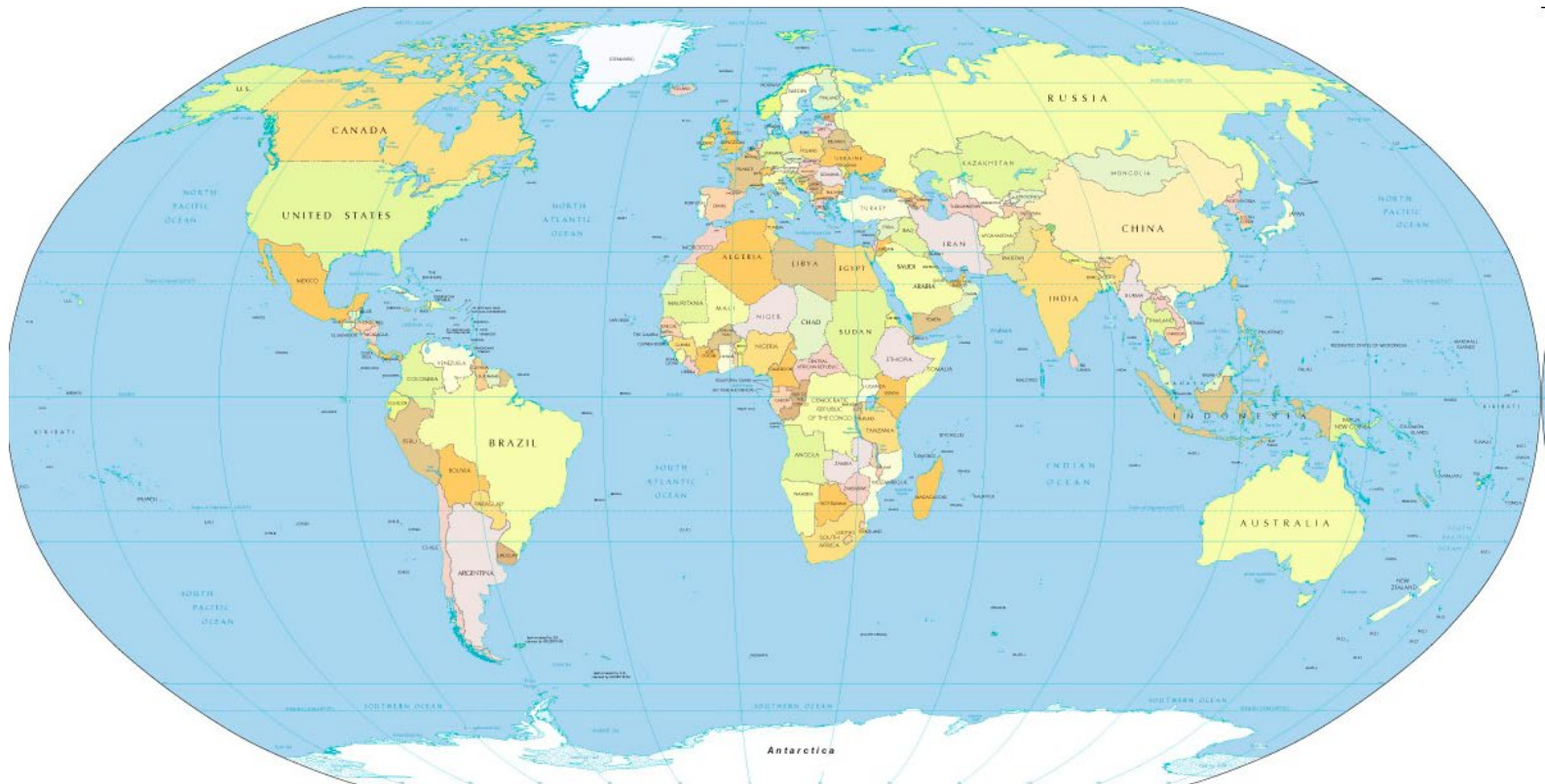
First steps – distance and dissimilarity measures

- How alike/different objects are
 - Species composition
 - Morphologically
- Dissimilarity measures (distance) more intuitive – can measure how “close” things in multivariate distance
- Usually presented in a dissimilarity matrix

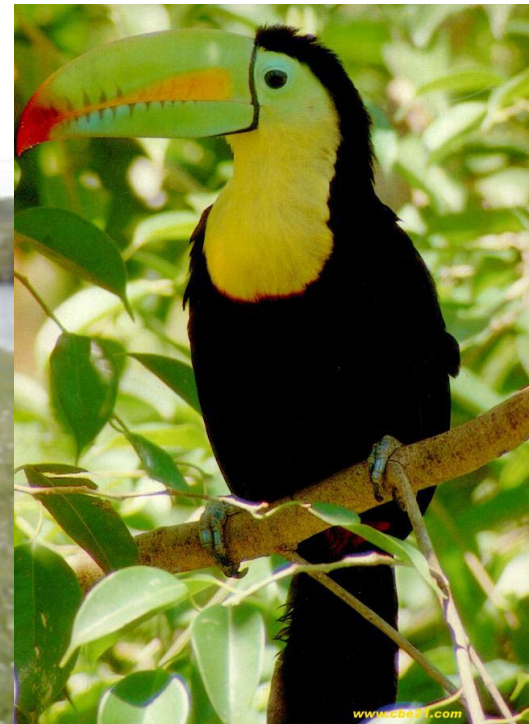
Measures of (dis)similarity

- Biological meaning when comparing pairs of samples
- Generally bounded by 0 and 100 (some have no upper limit and are standardised)
- 0 = similar, 100 = dissimilar, (hence “distances”)
- ***Problem of joint absences for ecologists - does it make sense to use a coefficient that depends on the number of species that are absent from both samples?***

- Antarctica and tropics - presence/absence reflect similarity
- *Joint absences* – if the a species is not present at both – Does this make the regions similar?



- Antarctica and tropics - presence/absence reflect similarity
- *Joint absences* – if the a species is not present at both – Does this make the regions similar?



Abundance array

Sample	1	2	3	4
Sp. 1	17	0	0	0
Sp. 2	21	0	0	13
Sp. 3	17	25	0	14
Sp. 4	0	119	135	117
Sp. 5	0	34	43	12
Sp. 6	0	0	0	7

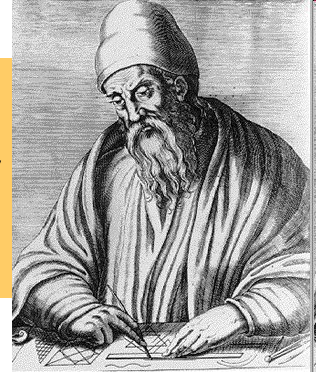
Which ones are most like each other?

e.g. Dissimilarity matrix

Sample	1	2	3	4
1	-			
2	74.4	-		
3	100.0	32.1	-	
4	47.8	31.9	58.0	-

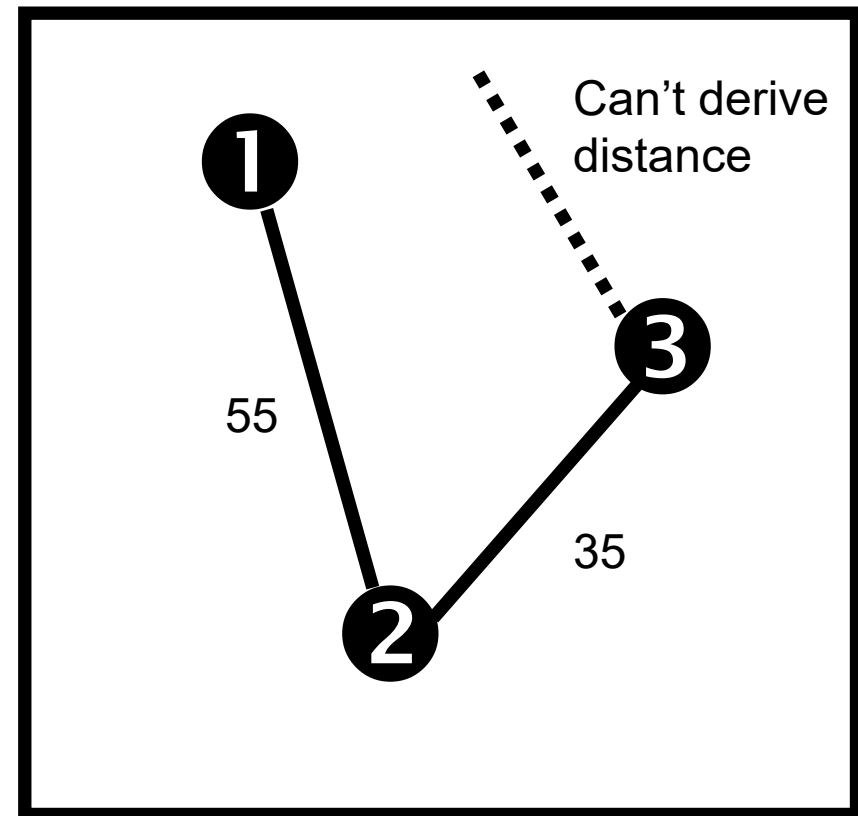
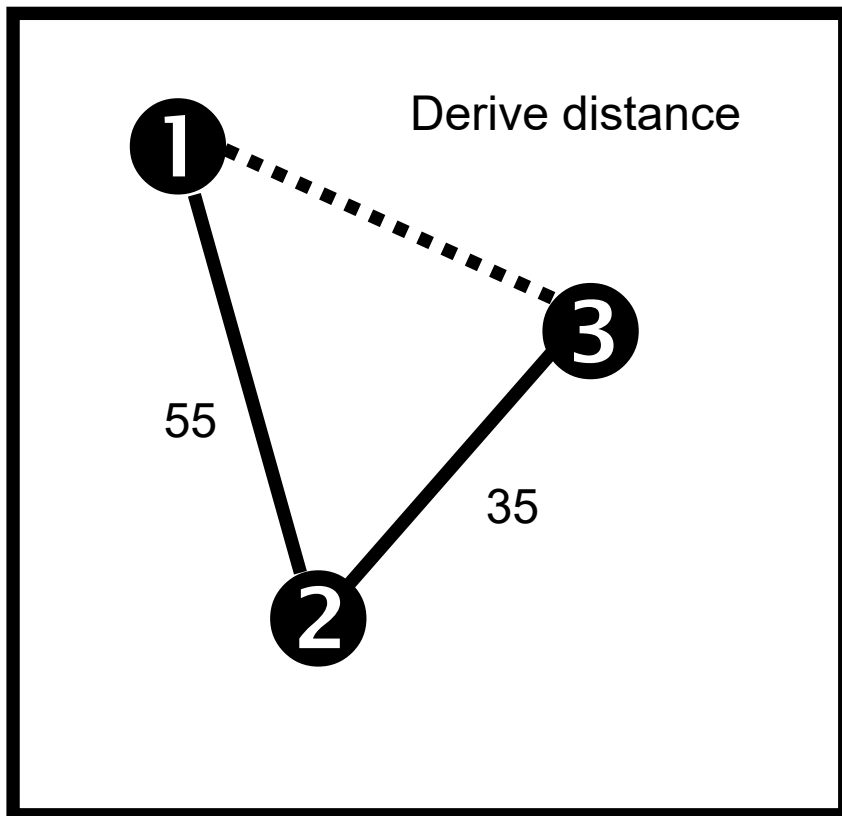
c.f. SIM matrix?

Options for (dis)similarity

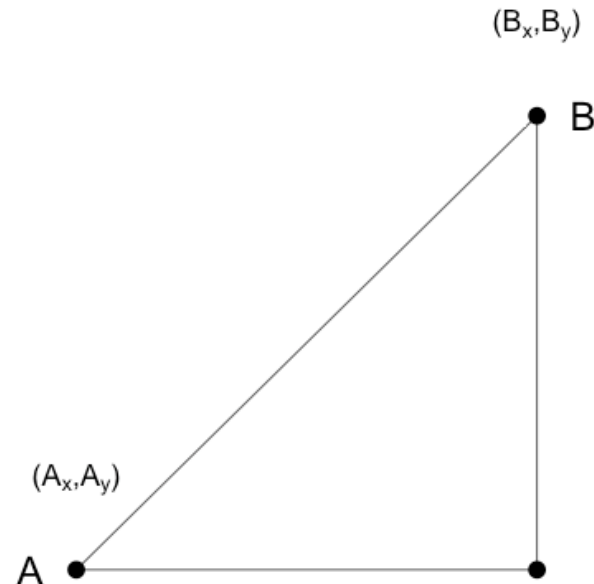


- Huge range to choose from
- Metric (continuous) vs non-metric (nominal, ordinal)
- Euclidean distance
 - geometry, square root of the sum of the squared differences between each variable
 - Useful for “measurement” variables with true zeros
 - Bounded by zero for samples with exactly the same values
 - no upper limit
 - Includes joint zeros
- c.f. Bray-Curtis one of the most popular distance measures in ecology

Metric vs. Non-metric measures



Euclidean Distance



The Euclidean Distance Between A and B =

$$\sqrt{(A_x - B_x)^2 + (A_y - B_y)^2}$$

Bray- Curtis dissimilarity

- a.k.a. percent dissimilarity
- Ignores 0/0 (joint absences)
- Main determinants are variables with high values
- Well suited to species abundance data
- Sum of differences between objects across variables standardised by sum of variable values across objects

Bray- Curtis dissimilarity

	Site 1	Site 2	Site 1-Site 2 (absolute)
A	15	10	5
B	8	10	2
C	4	0	4
D	10	5	5
E	4	6	2
Total	41	31	18

$$\text{Bray-Curtis} = \frac{\text{sum of absolute differences}}{\text{total abundances}} = \frac{18}{(41+31)}$$

How to calculate Bray-Curtis (dis)similarity in R?

```
# load package vegan
require(vegan)

# First calculate Bray-Curtis
dissimilarity matrix
bij1 <- as.matrix(vegdist(fossils))

# Then calculate Bray-Curtis
similarity matrix
BCI <- 100 - 100*bij1
```

Choosing dissimilarity measures?

- Metric v non-metric
 - An issue if used in linear models
- Presence of zeros
- Critical guide for choice of analysis
(suitability of analysis linked to dissimilarity measure)
- Pre-treatments - Standardisations and transformations

- Conventional transformations to help make relationships linear
- Important for analyses measuring linear relationships between variables
 - e.g. PCA



Transformation *(Clarke and Warwick 1994)*

- Absolute measures may over-emphasise the importance of prominent taxa
- Down-weighting importance of dominant species by changing scale of measurement
- Square root $\sqrt{y}$
- Log $(1+y)$
- 4th root $\sqrt[4]{y}$
- Presence/Absence



Abundance array

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Abundance array

Sample	1	2	3	4
Sp. 1	4.1	0	0	0
Sp. 2	4.6	0	0	3.6
Sp. 3	4.1	5.0	0	3.7
Sp. 4	0	10.9	11.6	10.8
Sp. 5	0	5.8	6.6	3.5
Sp. 6	0	0	0	2.6

Square root transformation $\sqrt{}$

Abundance array

Sample	1	2	3	4
Sp. 1	2.0	0	0	0
Sp. 2	2.1	0	0	1.9
Sp. 3	2.0	2.2	0	1.9
Sp. 4	0	3.3	3.4	3.3
Sp. 5	0	2.4	2.6	1.9
Sp. 6	0	0	0	1.6

Fourth root transformation $\sqrt[4]{}$

Abundance array

Sample	1	2	3	4
Sp. 1	1	0	0	0
Sp. 2	1	0	0	1
Sp. 3	1	1	0	1
Sp. 4	0	1	1	1
Sp. 5	0	1	1	1
Sp. 6	0	0	0	1

Presence/Absence transformation

Issues for transformations

- Cookbook approaches are well justified but you still need to make a case for your data
- Sometimes raw abundances of dominant variables mean something (e.g. water quality) – do you want to give undue weight to low range variables?
- Discretionary decision needs to be made with justification

Standardisation

- Adjust data so that means and/or variances or totals are the same for each variable
- Make each species equally important OR
- Make each sample equally important
- Express values as for each observation relative to the maximum (or total) value for that object, i.e. proportion of the largest value for that object
- Useful to compare things of different sizes (i.e. different area, different size)

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Standardise so that each sample is equally important

Standardise so that each species is equally important

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Standardise so that each sample is equally important

e.g. divide by the sum of each column, relative contributions of each species to sample

Abundance array

Sample	1	2	3	4
Sp. 1	0.31	0	0	0
Sp. 2	0.38	0	0	0.08
Sp. 3	0.31	0.14	0	0.09
Sp. 4	0	0.67	0.76	0.72
Sp. 5	0	0.21	0.24	0.07
Sp. 6	0	0	0	0.04

Standardise so that each sample is equally important

Sum=55

178

178

163

e.g. divide by the sum of each column, relative contributions of each species to sample

Abundance array

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e.g. divide by the sum of each row, relative contributions of each sample to species count

Standardize so that each species is equally important

Transformations and standardisations

- Making and justifying decisions
- “Arbitrary” – discretionary
- Recommendations? (from Clarke 1993, Quinn and Keough 2000)
 - Transformations – 4th root, preserves some abundance effects
 - Standardisations – change the fundamental interpretation, use in addition to “raw” data?
- Look at your data and your question

Take home points (part 1)

- Basic vocabulary for multivariate approaches
- How to choose distance measures - fundamental principles
- Transformations and standardisations
 - Why you would do it
 - Different types
 - How they affect our interpretation of results

Take home messages (part 2)

- Understand the process – from distance (or similarity) matrix to analyses
- Decision making and options
- Merits of different approaches
 - Rationale
 - Pros and Cons
 - Interpretation of outputs

Using similarity matrices

- Cluster analysis
 - Types of questions
 - Carrying it out
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